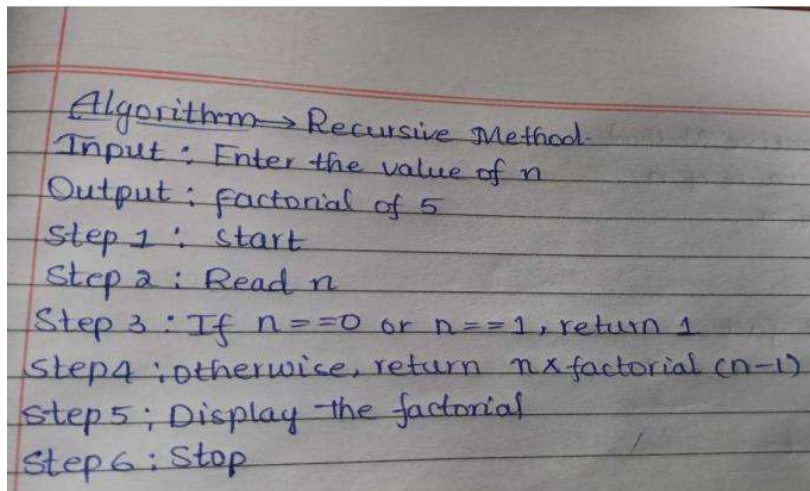


PRACTICAL-4

Aim: Implementation and Time analysis of factorial program using iterative and recursive method.

Algorithm:



Program:

1) Recursive method

```
#include <iostream>

using namespace std;

class Factorial
{
public:
    int factr(int n)
    {
        if (n == 0 || n == 1)
        {
            return 1;
```

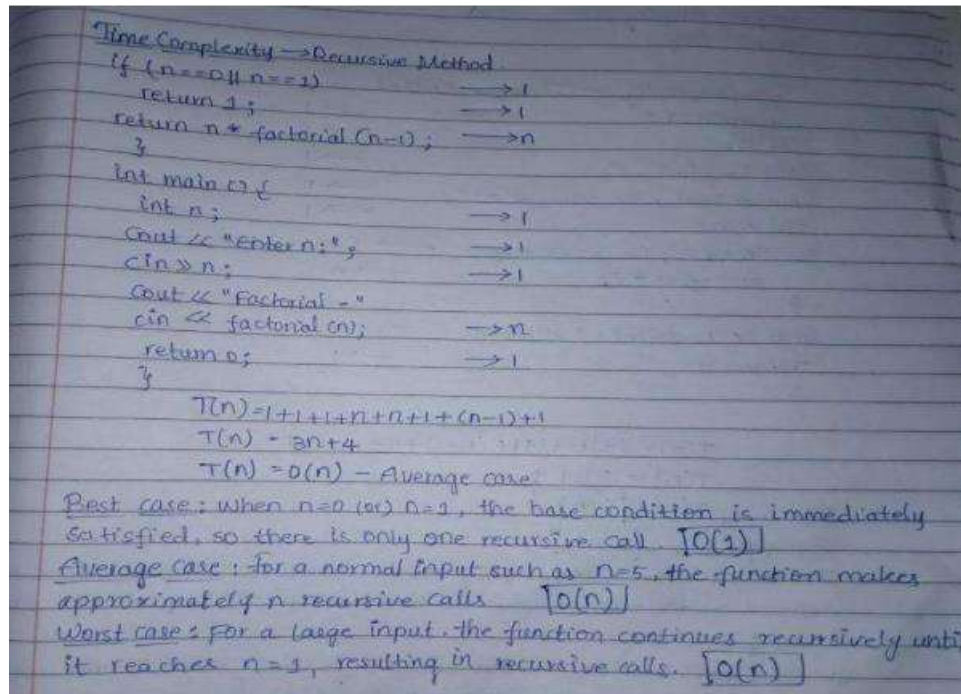
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```
    }  
    else  
    {  
        int result = n * factr(n - 1);  
        return result;  
    }  
}  
};  
  
int main()  
{  
    int n, result;  
    // Create an object of the class  
    Factorial obj;  
    cout << "Enter the value to find its factorial: ";  
    cin >> n;  
    result = obj.factr(n);  
    cout << "Factorial of " << n << " is " << result << endl;  
    return 0;  
}
```

Output:

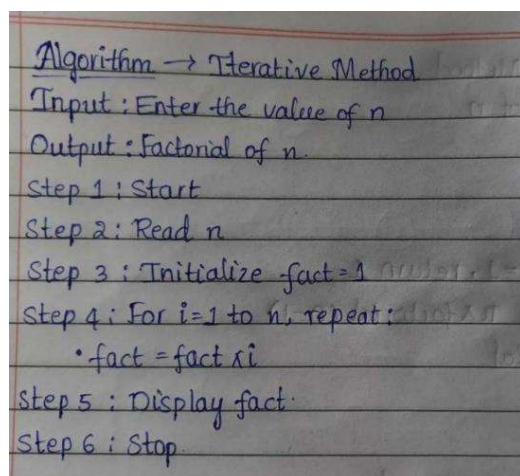
```
Enter the value to find its factorial: 12  
Factorial of 12 is 479001600  
  
Process returned 0 (0x0)    execution time : 3.113 s  
Press any key to continue.  
|
```

Time complexity:



2) Iterative method

Algorithm:



Program:

```
#include <iostream>

using namespace std;

class Factorial
{
public:
    int fact(int n)
    {
        int result = 1;

        for (int i = 1; i <= n; i++)
        {
            result = result * i;
        }

        return result;
    }
};

int main()
{
    int n, result;

    Factorial obj;

    cout << "Enter the value to find its factorial: ";

    cin >> n;

    result = obj.fact(n);

    cout << "Factorial of " << n << " is " << result << endl;
}
```

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```
return 0;  
}
```

Output:

```
Enter the value to find its factorial: 12  
Factorial of 12 is 479001600  
  
Process returned 0 (0x0)   execution time : 2.985 s  
Press any key to continue.
```

Time complexity:

Time Complexity $\rightarrow$ Iterative Method

```
int main() {  $\rightarrow 1$   
    int n;  $\rightarrow 1$   
    long long fact = 1;  $\rightarrow 1$   
    cout << "Enter the value of n:";  $\rightarrow 1$   
    cin >> n;  $\rightarrow 1$   
    for (int i = 1; i <= n; i++) {  $\rightarrow 1 + (n+1) + n$   
        fact = fact * i;  $\rightarrow n$   
    }  
    cout << "Factorial = " << fact;  $\rightarrow 1$   
    return 0;  $\rightarrow 1$   
}
```

$T(n) = 1 + 1 + 1 + 1 + (n+1) + n + n + 1 + 1$
 $T(n) = 3n + 9$

Best case: $T(n) = O(n)$
Average case: $O(n)$

In the iterative factorial program, the number of iterations does not depend on the type or value of the input n , the loop always executes exactly n times.

Worst case: The loop always executes n times, even in the worst case termination. $O(n)$.

Conclusion:

The factorial program was successfully implemented using both iterative and recursive methods. The iterative method has $O(n)$ time and $O(1)$ space complexity, while the recursive method has $O(n)$ time and $O(n)$ space complexity due to the recursive call stack.